Can a statistics machine pass the German state examination?

– Replicating LEXam on German First State Exam-style essay cases to probe the “statistical” face of legal reasoning –

LEXam-DE · A benchmark experiment

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1,127 German exam cases · 14 models under the LEXam protocol + 12 systems under the German rubric · 37 human participants · ~23,000 newly graded answers

The question

If a statistics machine can pass our exams, what does that say about the exams?

- **Never taught legal method.** Language models predict text. Yet they write fluent Gutachten with Obersatz, definition, subsumption, and conclusion.
- **Two possible answers:**
  - **Imitation:** the machines only copy the *surface* of legal argument. Then a careful grader must catch them.
  - **Pattern:** parts of what we grade, such as structure, phrasing, and the form of the Gutachtenstil, are *regular patterns* that statistics can learn.

→ **Both answers are informative. We measured it.**

Two benchmarks - one goal

The experiment: we replicated LEXam faithfully on the BenGER corpus. **One corpus, two complete evaluation pipelines**, each with its own prompt and grader. The two evaluation philosophies can be compared at the system level.

LEXam (Fan et al. 2026)

THE VALIDATED FOUNDATION

- 340 University of Zurich law exams. The reference methodology for LLM legal-exam evaluation
- An LLM examiner scores each answer **against the reference solution** (0–100). Validated against human legal experts
- Lean, strict, and reproducible. Substance only. Deviations are penalised

BenGER (Nagl et al. 2026)

THE GERMAN INSTANTIATION

- 1,127 German cases with official Musterlösungen, curated by legal experts on an open platform
- Graded as a German corrector would. **100-point doctrinal rubric mapped to Notenpunkte** (~70 substance, ~30 craft)
- Human-anchored. 37 participants wrote the same exams. Student cohorts serve as calibration

Setup - corpus and models

Three difficulty tiers, fourteen models, no help.

- **Corpus:** 1,127 German cases with official Musterlösungen
  - **Grundprinzipien:** 531 short yes/no principle questions (civil & criminal law)
  - **Benchathon:** 15 curated exam cases, answered by **37 humans** (220 solutions)
  - **ZJS cases:** 581 full practice exams in state-exam format. Reference solutions average 30,000 characters
- **Models:**
  - **14 affordable-tier models** from every major provider, including the 3 with published Swiss LEXam scores
  - **Flagships** (GPT-5.4, Claude Opus, Gemini Pro) join the rubric analysis
  - **Zero-shot:** LEXam's prompt, byte-identical up to the Swiss-to-German substitutions. No system prompt, no examples, no retrieval

→ **Fully reproducible: prompts, configurations, and analysis code are published. The platform is open source.**

Replication - kept and changed

The replication worked. Same qualitative ordering among the shared anchor models. 0 judge errors over 23,000 cells. Total cost under €100.

- **Kept exactly:**
  - The LEXam **generation prompt, verbatim** (only Swiss law to German law)
  - The LEXam **judge prompt** and its 0–1 correctness scale vs. the reference answer
  - **Zero-shot:** no system prompt, no examples
- **Changed (engineering, not method):**
  - **One judge model** instead of the 3-judge ensemble, to save cost. Rankings stay comparable, levels read slightly milder
  - **Structured JSON** judge output instead of free text
  - **16k token caps**, never binding (<1% truncation)
  - **MCQ track adapted to yes/no** (our corpus has no MCQ data)

→ **The LEXam recipe travels. That is a credit to how it was built.**

Three graders

The core idea: where do the three graders agree, and where do they part ways?

1 · Correctness judge (LEXam's method)

An LLM examiner compares the answer with the reference solution and scores substantive fulfilment from 0 to 100. It rewards substance and penalises deviation.

"Is it right?"

2 · German doctrinal rubric (BenGER)

A 100-point scheme mirroring German correction practice (result, completeness, subsumption, weighting, structure, language), converted to **Notenpunkte**.

"What grade would a German corrector give?"

3 · Similarity metrics (NLP)

They measure only how *similar* the answer is to the reference, by shared words or computed meaning closeness. They know nothing about law.

"Does it sound like the reference solution?"

Full exams - the German rubric

581 real practice exams, graded under the German doctrinal rubric. **The field splits at the flagship line.**

7.3 grade points

best system average (Gemini-3.1-Pro), a solid "befriedigend" on full state-exam-style Klausuren

76–83%

of flagship exams reach "ausreichend" or better (pass bar: 50/100 ≈ grade 4)

5–30%

pass rate of the weakest open-weight systems (2.4–3.7 grade points)

2–6

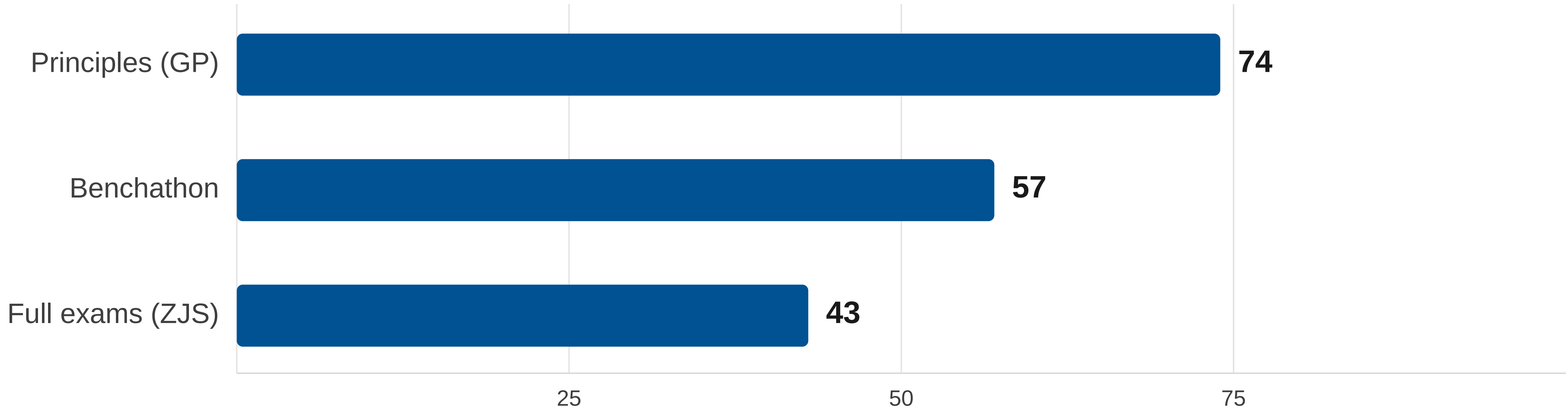
grade points: real student cohorts on comparable practice exams (anonymised, same platform)

Flagships score above a typical practice cohort. Everything below that tier largely fails. Source: BenGER benchmark (Nagl et al. 2026).

The difficulty gradient

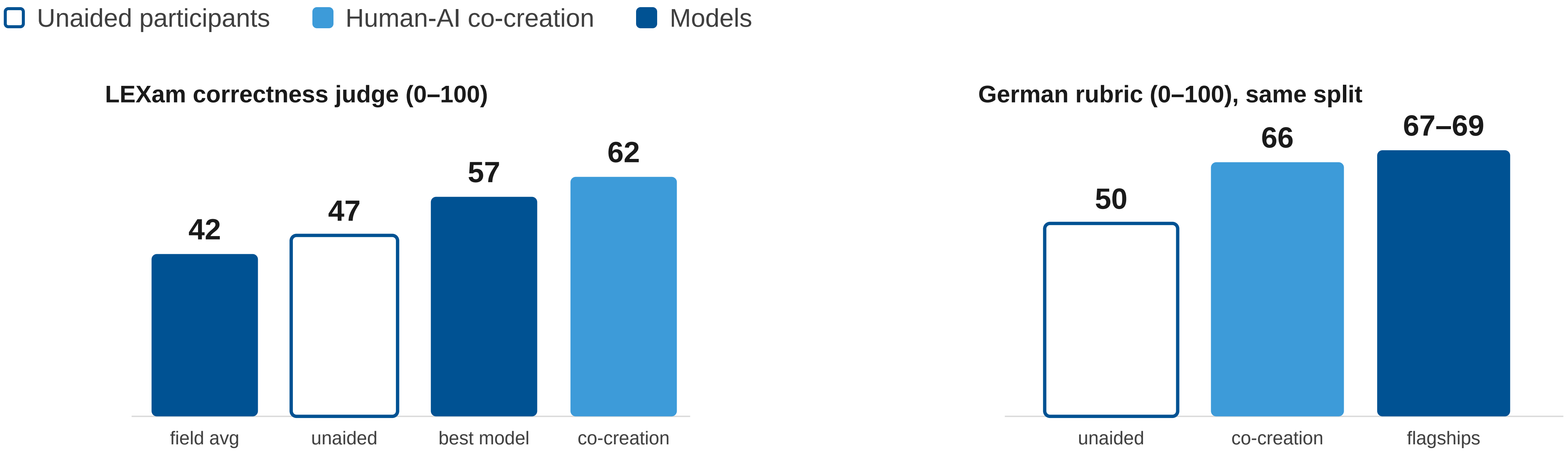
Competent on principle questions, overwhelmed by the full exam. The models have absorbed a great deal of German doctrine. But none of them sustains a complete, correctly weighted solution for hours. That is the core competence the state exam certifies.

Best model per tier, correctness judge (0–100). The ZJS field stretches down to 22:



Humans vs. machines

Co-creation leads under both graders. Unaided work sits mid-field. The **~15-point co-creation uplift** appears under both instruments (+14.9 judge, +15.7 rubric). Unaided participants beat the affordable field's average but not its leaders.

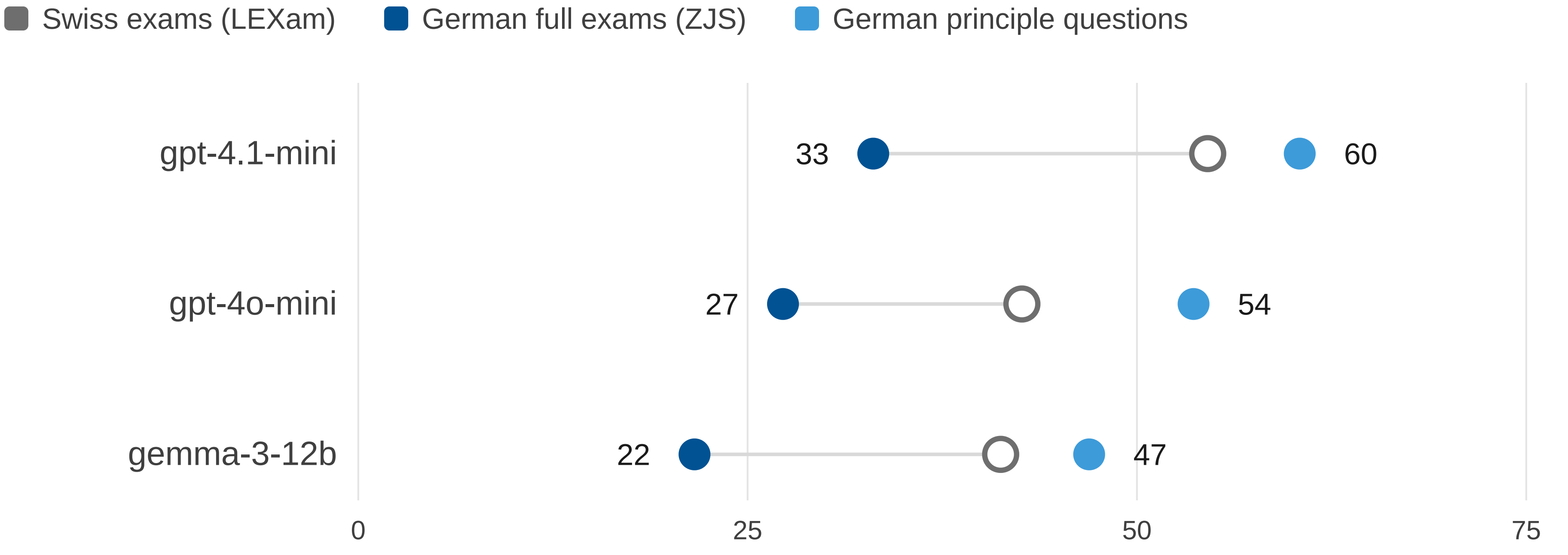


Left panel: affordable tier only. Humans vs. flagships under the LEXam judge remains open. Source: BenGER benchmark (Nagl et al. 2026).

Germany vs. Switzerland

The format is the hurdle, not the language. The three models with published Swiss scores land **15–22 points lower** on German full exams and **5–11 points higher** on short principle questions. What they cannot sustain is the disciplined structure of the complete Gutachten.

Three models with published Swiss LEXam scores, identical method and grading logic:

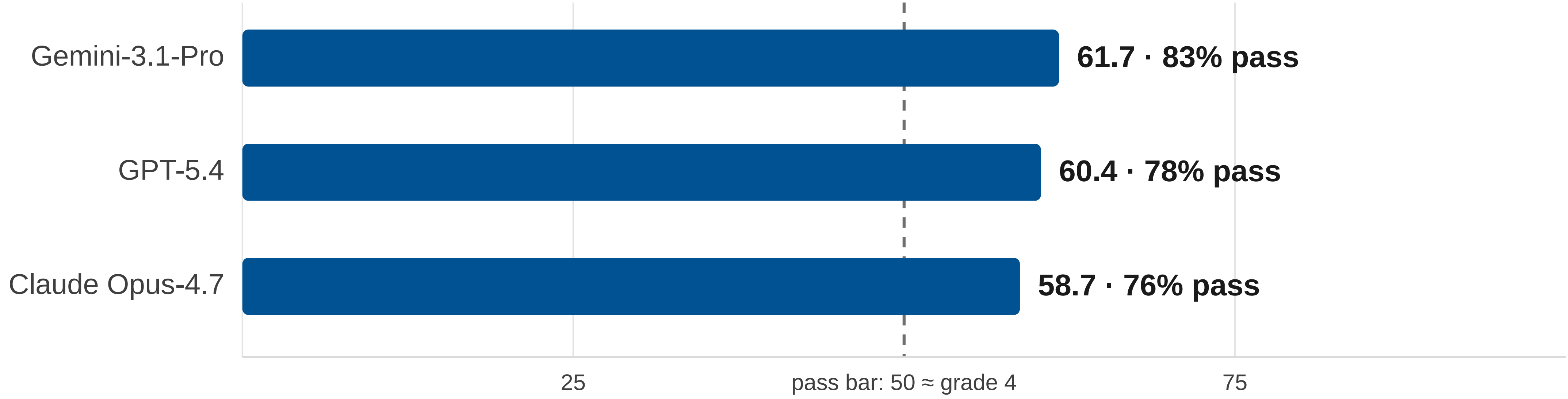


Swiss scores: LEXam's stricter 3-judge ensemble. Ours: one milder judge. The exam deficit is conservative, and part of the principles gain may be leniency.

The flagship tier

Genuinely capable: a solid "befriedigend" on full exams (6.5–7.3 grade points), and up to 9.6 grade points with ≥93% pass on the shorter Benchathon cases. **The ceiling is real:** no system's average approaches the distinction-level grade bands.

Flagship systems, ZJS full exams, German rubric (0–100), BenGER benchmark (Nagl et al. 2026):



Why not just LEXam on German data?

Not a replacement but an extension. Replicating LEXam gave us a validated external yardstick. That yardstick makes the whole comparison possible. But the German exam asks questions a correctness score is not designed to answer:

- **Format:** the multi-hour Gutachten makes craft (Gliederung, method, language) part of official German grading practice. A reference-based score *rightly* excludes craft. A benchmark for German legal education must include it (~30/100 rubric points).
- **Vertretbarkeit:** German correction credits defensible alternative solution paths. A single-reference anchor penalises them. About 6.5% of answers sit in exactly this spot.
- **Human anchoring:** expert curation and human baselines on a living platform give every machine number a human reference point.

Cross-check - two graders, one ranking

In short: BenGER exists alongside LEXam rather than instead of it. One number strips form away. The other shows what a German corrector would actually write.

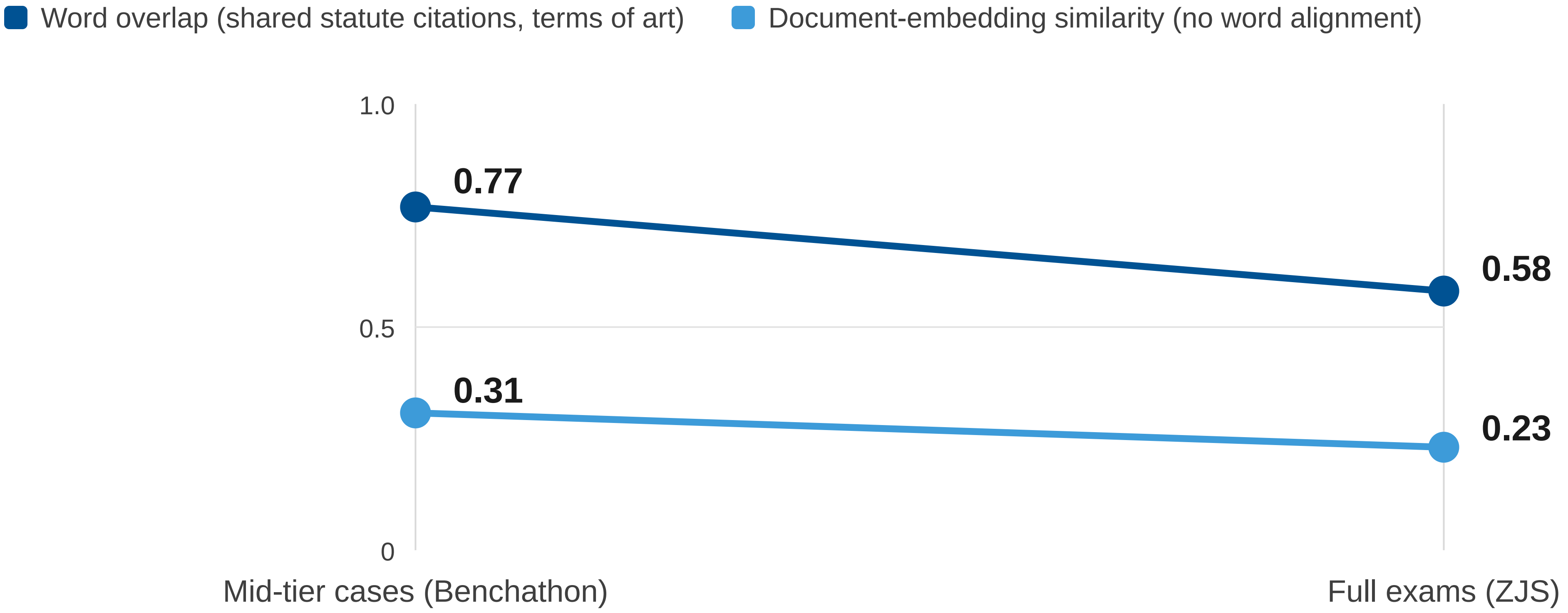
- **The rankings agree:** across 11 models, the correctness judge and the rubric produce nearly identical rankings. One adjacent pair is swapped.
  - **For LEXam:** the lean recipe survives a jurisdiction change and a much longer exam format
  - **For BenGER:** the rubric's substance core moves with the established method
- **The levels differ:** the rubric grades 1–10 points higher, and the gap grows with model strength. Consistent with fluent models collecting most of the **~30 craft points** the correctness judge ignores.
- **The second agreement:** the co-creation uplift measures **~15 points under both instruments** (+14.9 judge, +15.7 rubric). Two independently built graders detect the same effect.

→ **The graders agree on which machine is better. They disagree on how much fluent form is worth.**

Similarity vs. grade

Similarity predicts the grade, but less well on harder cases. On easy cases, sounding like the reference solution and deserving a good grade are nearly the same thing. On hard exams they are not.

Rank agreement between similarity-to-reference and rubric grade (1.0 = perfect, 0 = none):



A third or more of the grading signal is invisible to every similarity measure. Over 10,000 paired gradings on ZJS alone.

Where similarity and grade come apart

5.0 %

Sounds like the reference, fails anyway. Example: a Claude Opus answer in the top similarity third with 29/100 points. It paraphrases the terrain but omits the decisive doctrinal steps.

35.7 %

Aligned: similar and good, or dissimilar and weak. Here statistics works as a grader.

52.8 %

Middle terciles

6.5 %

Sounds different, but is good. Consistent with legitimate alternative paths (vertretbar). Similarity punishes what doctrine rewards.

All ZJS answers, split into thirds by similarity and by grade.

The outliers mark exactly what similarity *cannot* see: the decisive step, the weighting, recognising what the case turns on.

That is also where the machines fail most often.

Three consequences

- **For legal education:** flawless structure is now cheap, for every student and every machine. The discriminating signal has moved into issue-spotting, weighting, and the decisive doctrinal step. Exam design and point schemes should follow it there.
- **For the profession's self-understanding:** part of legal argument is demonstrably pattern-like. Machines master it, and statistics can even *grade* it. The rest is identifiable, teachable method. The distinction-level grade bands it guards remain human territory for now.
- **For benchmarking:** the field does not need one benchmark to win. It needs a common validated foundation (LEXam) and jurisdiction-specific instruments on top of it (BenGER). *Whoever configures the grader defines legal quality.* That demands the transparency of human examination regimes.

Method and limits

The caveats do not touch the central pattern. Capability is tiered. The human baseline splits by working mode, not by grader. And there is a measurable gap between what similarity sees and what doctrine demands.

- **Reproducible and cheap:**
  - Prompts, configurations, and analysis code are published. The corpus is the published **BenGER dataset**. The platform is open source
  - Total cost of the experiment: **~€75**
- **Limits:**
  - The **graders are themselves language models**. LEXam validated its judge against human legal experts. Residual biases remain possible
  - **One judge model** instead of LEXam's ensemble, to save cost. Rankings comparable, levels somewhat milder
  - **Flagships appear only in the rubric pipeline.** Humans vs. flagships under the LEXam judge remains open
  - One provider **refused a criminal-law fact pattern** on content-policy grounds. That is itself a finding about AI in legal use
  - **15 cases and 220 human solutions** in two working modes. A small but clean sample

what-a-benger.net

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